



Rethinking the choice of carbon tax and carbon trading in China

Zhijie Jia, Boqiang Lin*

School of Management, China Institute for Studies in Energy Policy, Collaborative Innovation Center for Energy Economics and Energy Policy, Xiamen University, Fujian 361005, PR China



ARTICLE INFO

Keywords:

Carbon tax
Carbon cap-and-trade
Emission trading scheme
Computable general equilibrium (cge) model
emission reduction

ABSTRACT

Carbon tax and carbon emission trading are used as emission reduction strategies. This paper re-analyzes the differences between the carbon tax and carbon trading by applying a recursive dynamic computable general equilibrium model, called the CEEEA (China Environmental-Energy-Economy Analysis) model. Unlike previous literature, we make the gross domestic product an exogenous variable and explore different effects between carbon tax mechanism and carbon trading mechanism on the environment, energy, and economy. Given a constant GDP effect, both carbon trading and carbon tax have strong emission reduction capacity, but the relative emission reduction efficiency of the carbon tax is higher than that of carbon trading. This advantage increases over time. Carbon trading has a certain negative effect on the output of the energy industry, and also on the output of other energy-intensive industries involving in carbon trading. After considering the various invisible costs of establishing a new carbon trading market, this paper recommends that China could directly levy a carbon tax on energy enterprises, or just increase the production tax on fossil fuels to reduce CO₂ emissions effectively. In this way, the commodity market can be used for price incentives and thus, achieve the goal of emission reduction.

1. Introduction

1.1. Background

It is generally believed that the main cause of global warming is the excessive carbon dioxide emissions as a result of the huge demand for fossil energy by human activities (Xu and Lin, 2018a). Carbon dioxide is one of the greenhouse gases. Reducing carbon emissions, promoting a low-carbon economy, and implementing low-carbon economic actions have become a consensus for most countries and an inevitable choice of human development. Therefore, the smooth and rapid transition to a low-carbon economy has become a topic of interest (Xu and Lin, 2018b). Specifically, how to use emission reduction tools to achieve emission reduction goals has become one of the important aspects of emission reduction studies (Du et al., 2019; Higgins et al., 2011).

There are two main policy strategies for carbon dioxide emission reduction. Firstly is the command and control regulation, which is relatively rigid and unable to play the role of a competitive market. The other is based on price incentives, which is more conducive for a competitive market mechanism. Strategies for price incentives can be divided into the means of total amount control and the means of price control. The emission trading scheme (ETS) is a typical representative

of the total amount control method. Its typical design scheme is cap and trade (Chan and Morrow, 2019). ETS is to re-establish a carbon emission trading market. The trading of scarce carbon emission allowance is a manifestation of the market, indicating the role of the market and the improvement of emission reduction efficiency. The carbon tax (CT), energy tax and other tax methods are typical representatives of the price control method for energy production or consumption enterprises (Zhang et al., 2019b). A tax mechanism that is based on a carbon tax or other environmental taxes, is a way of price control (to directly increase the energy consumption cost). Compared with the total amount control, the tax mechanism is easier to understand and implement but hard to explore the real effect (Lin and Jia, 2018).

Many studies are focusing on the comparison of CT and ETS, however, few studies compare different emission reduction strategies at an impact level, e.g. the same GDP impact or emission reduction effect. This is the knowledge gap that we want to fill.

1.2. Previous studies and motivations

The carbon tax is a kind of consumption tax based on the carbon content or carbon dioxide emission of fossil fuels. According to Pearce (1991), the carbon tax has the effect of "double dividend": it does not only improve the environmental quality but offsets the income

* Corresponding author.

from carbon tax against other corporate taxes increase employment and investment, and then, makes the economy more efficient. Researches on carbon tax have continued to these days. Such as, [Frey \(2017\)](#) used the computable general equilibrium model to analyze the impact of different carbon tax levels on Ukraine's economy and environment. The effects of carbon tax are widely explored by researchers which is similar to the above literature, such as the effect in Spain ([Forero-Cantor et al., 2020](#)), South Africa ([Nong, 2020](#)), China ([Ma et al., 2020](#); [Tan and Lin, 2020](#)), India ([Bhat and Mishra, 2020](#)), and Britain ([Kong Chyong et al., 2020](#)), etc.

The efficiency of a carbon tax is researched extensively. Carbon tax can somehow motivate enterprises to invest on energy efficiency and finally reduce long term carbon emissions ([Dorsey-Palmateer and Niu, 2020](#)), as well as the behavior of other consumers, such as households. [Hájek et al. \(2019\)](#) tested the efficiency of a carbon tax in the energy industries of the EU and made a positive conclusion. A similar positive conclusion in emission reduction effect and efficiency of a carbon tax can be found in many pieces of research, such as [Ntombela et al. \(2019\)](#) and [Olale et al. \(2019\)](#). Moreover, [Andersson \(2019\)](#) argued that the emission reduction effect of CT may be underestimated by using price elasticities for simulation. Although the efficiency of a carbon tax is nearly common sense, there are many hints indicating why carbon tax is replaced by ETS in many regions. Australia experienced incredible carbon policy changes within a short time frame during 2009 and 2014 - (1) the ETS move, (2) the carbon tax, and (3) the repeal of the carbon tax. The lengthy processes of climate policy debate and legislative changes limit the value-enhancing capability of carbon tax performance ([Qian et al., 2020](#)). The uncertainty of carbon tax policy will encourage emitters to delay their energy efficiency investment, while waiting and seeing if the carbon tax policy will last. And the public attitudes towards carbon tax is getting bad, especially when people see the benefit opportunity in ETS market ([Umit and Schaffer, 2020](#)).

The distribution effect of carbon tax controversial. In general, carbon tax may cause a retrogressive effect. Carbon tax is usually retrogressive in many developed regions, while it may reveal progressivity in developing regions sometimes ([Wang et al., 2016](#)). In many studies, the distribution effect varies, such as, it shows a slight progressivity in Mexico ([Renner, 2018](#)), but the distribution effect of the carbon tax is significantly regressive in Shanghai ([Jiang and Shao, 2014](#)), and slightly regressive in whole China in terms of rural and urban household or between provinces ([Wang et al., 2019](#); [Zhang et al., 2019a](#)). Another, more researchers believe that the revenue recycling scheme affects distribution effect ([Vandyck and Van Regemorter, 2014](#)). Carbon taxes are not necessarily regressive, rather, the way revenue is recycled is a major determinant of how the carbon tax costs are distributed ([Gonzalez, 2012](#)). There are many researches on distribution effect of carbon tax. [Yamazaki \(2017\)](#) explored the impact of a revenue-neutral carbon tax on employment in British Columbia in 2008 and showed that the revenue-neutral carbon tax may not adversely affect employment. [Fremstad and Paul \(2019\)](#) argued that inequalities still exist even when there are labor tax cuts in the context of a carbon tax. [Rosas-Flores et al. \(2017\)](#) analyzed the distribution effect of subsidies and a carbon tax on Mexican households. Without sufficient revenue recycling, the carbon tax is regressive ([Berry, 2019](#); [Saelim, 2019](#)).

Previous studies on emission trading are quite different from that of the carbon tax since emission allowances have financial attributes. However, many other pieces of research focused on the effects and efficiency of ETS ([Lin et al., 2019](#)). ETS is one kind of environmental regulation. The stable and long-term regulation follows Porter hypothesis ([Liu et al., 2020](#)). ETS does reduce emission by changing the behaviors of energy users ([Chao et al., 2019](#)), improve energy efficiency to some extent ([Zhang and Fan, 2019](#)), and have different reduction effects based on different ETS plan ([Chang et al., 2016](#); [Mu et al., 2018](#)). The effect of ETS on reducing emissions is also significant, such as, China's ETS pilot does have a significant 'reduction effect' on haze

pollution concentration level ([Yan et al., 2020](#)). The cost of emission trading is also considered by many pieces of these researches. Such as the negative impact on the GDP. The impact of ETS on GDP in China ranges from 0.19% to 1.44% under different schemes ([Lin and Jia, 2019](#)). GDP in Vietnam will reduce by 4.57% if restricting sectorial coverage and 1.78% in if full coverage ([Nong et al., 2020](#)). In many cases, cost and effectiveness are trade-offs when we talk about ETS ([Craig et al., 2017](#); [Peng et al., 2019](#); [Zhang et al., 2020](#)), as well as carbon tax ([Goulder et al., 2019](#)).

The comparison between emission trading and a carbon tax is also examined by a certain researchers. [Strand \(2013\)](#) analyzed different effects of the carbon tax and emission trading from the perspective of international trade, offsets, etc. [Barragán-Beaud et al. \(2018\)](#) compared the two emission reduction tools from the perspective of political feasibility and policy mechanism and concluded that ETS is the most appropriate tool in Mexico. A similar result is found in the Chinese remanufacturing industry ([Hu et al., 2020](#)). The given schemes of carbon trading or carbon tax are compared in previous literature, but there is a lack of a benchmark. Previous studies failed to compare them at the same economic level or at the same level of reduction effect. This is the gap we want to fill to provide an accurate comparative analysis.

As it is not objective to analyze the impact of two policies on GDP and emission reduction efficiency simply by comparing the assumed carbon tax and carbon trading. Therefore, this paper will adjust the carbon tax mechanism and carbon trading mechanism to let the 2030 carbon tax and carbon trading policy affect GDP equally. Then compare the advantages and disadvantages of the two emission reduction strategies, determine the best strategy, which is more suitable for China, and discuss the policy implications.

To do such counterfactual analysis, traditional econometric paradigm has the limited power and so do engineering analysis like Data Envelopment Analysis. In this case, various models of equilibrium analysis based on game theory have strong advantages in simulating counterfactual scenarios. Among these models, Computable General Equilibrium (CGE) model can simulate the different subjects in society, such as subject, producers, consumers, government, and foreigner etc., and the relationship among them. And more importantly, most data used by CGE model is from input output table, which means CGE model could easily describe the different behaviors in different kinds of enterprises based on actual data. On the topic of comparison between CT and ETS, we need to find the different impact on these industries. So CGE model is the chosen one for us to analyze this issue.

1.3. Contribution and structure

The paper's main contributions are as follows:

- 1 Based on previous literatures, we provide a more objective comparison between two policies: given the same impact of ETS and carbon tax on GDP, we analyze the impact of ETS and CT on China's economic & social indicators. With the same benchmark, the comparison between carbon tax and carbon trading will be more objective. This is a contribution from the method of research.
- 2 This paper provides a detailed modeling framework and equations, which can be used as a reference for CGE modelers analyzing energy or environmental problems in the future. This is another contribution from the method of research.
- 3 In consideration of key conclusion of this paper, we put forward a series of implications and suggestions on low carbon policy pathway by assuming that China's actual ETS situation changes. The suggestions are very targeted. This is a contribution from the perspective.

The rest of the paper is structured as follows. [Section 2](#) mainly introduces the study methods: the CGE model, the social accounting matrix, and the data source. [Section 3](#) discusses the design of the carbon

tax and emission trading scenarios. Section 4 provides the research results. Section 5 presents the conclusion and policy implications.

2. Methodology

2.1. CEEEA/CGE model

CGE models have been widely used for policy analysis and simulation. This paper applies a dynamic recursive computable general equilibrium model, the China Environmental-Energy-Economy Analysis (CEEEA) model. Like other CGE models (Li et al., 2017), the structure of the CEEEA model in this paper is based on the traditional Walras paradigm. The model describes the process of all subjects (residents, governments, enterprises, etc.) seeking to maximize their objective functions simultaneously. The CEEEA model in this paper is composed of five blocks: production block, income and expenditure block, trade block, carbon tax and carbon trading block, and the macro closure and market clearing block (see Lin and Jia (2019) for details of other models).

2.1.1. Production block

The production block describes the production process of the enterprise, as shown in Fig. 1. The CEEEA model assumes that each sector only produces one kind of product. The products produced by the sector are homogeneous. Enterprises use different kinds of factors and intermediate inputs for domestic output. We assume that domestic output is composed of VAE (value-added and energy bundle) and intermediate input using the Leontief production function. This is the top level of the production block. The VAE consists of value-added (VA) and energy. It follows the constant elasticity of substitution (CES) production function. VA consists of capital input and labor input and follows the CES production function. The energy bundle consists of electricity input and fossil energy input and uses the CES production function. Fossil energy input consists of coal and oil & gas, which follows the CES production function. In this production block, it is important to note that: 1) the intermediate input is the intermediate input excluding the energy industry and the energy demand from each sector as reflected in the energy input bundle. 2) Because the input-output table of China, with 139 sectors, does not differentiate the oil and gas industries, similarly, the CEEEA model does not divide the energy input of the oil and gas industries. However, since coal consumption accounts for more than 60% of primary energy consumption in China, the results are not affected by this aggregation.

2.1.2. Income and expenditure block

As shown in Fig. 2, the income and expenditure block describes the economic relationship between the economic entities in the society; the government, domestic enterprises, urban residents and rural residents. Except for the government, other subjects maximize their reward functions: domestic and international enterprises pursue profit maximization while the two types of residents pursue utility maximization.

Therefore, the behavior preference of the economic subjects described in this paper is determined by the maximization of self-pursuit. Through the sale of labor and capital, residents get revenue from enterprises and also get transfer payments from the government. All income is used for consumption, savings and direct tax. A domestic enterprise makes its income by selling its products. The income is used for purchasing inputs (intermediate inputs), payment of remuneration to residents, payment of indirect taxes and savings. The government obtains income through direct tax, indirect tax, and tariff. All the income is used for government consumption, government savings (deficit) and transfer payments. Foreign enterprises sell goods produced abroad to domestic residents, enterprises, and governments.

2.1.3. Trade block

Fig. 3 shows the framework of the trade block. Most CGE model assumes that the products produced by the same sectors are identical, that is, the products produced by the same industry are homogeneous. This assumption implies that the commodity can only be imported or exported in the trade block. However, countries could trade the same commodity. Therefore, like some CGE studies, this paper introduces the Armington assumption. This means that the residents and enterprises in the model are not directly consuming imported commodities or domestic output, but consuming the so-called Armington commodities. Armington commodities are composed of imported and domestic products. This paper further assumes that domestic enterprises need to determine the proportion of supply to the domestic market and the international market. We use the constant elasticity of transformation (CET) function to describe the conversion process.

2.1.4. Carbon tax and carbon trading block

Currently, at least 20 countries in the world have imposed carbon taxes. These countries can be divided into three categories: those with a relatively complete carbon tax system, strong policies, and early implementation. Examples are Denmark, the Netherlands, etc.; the second category adopts carbon tax under the global joint emission reduction with weak implementations. Examples are Australia, France. The third category is those not actively implementing carbon taxes to reduce carbon emissions. Except for the tax rate of a carbon tax and the coverage of industries, other carbon tax mechanisms are based on the system of Denmark, the Netherlands and other countries in the first category: ① the carbon tax rate is fixed and paid in the form of energy tax; ② for the industries that paid carbon tax, the carbon tax can be used to offset the value-added tax and other taxes. Up to 50% of the indirect tax can be offset. Also, this paper assumes that the carbon tax is levied on the energy production enterprises. The carbon tax design equations are:

$$EM_i = COAL_i \times \gamma^{coal} + O_G_i \times \gamma^{o-g} \quad (1)$$

$$PLC_{en} = p^{\alpha} EM_{en} \quad (2)$$

Where EM_i is CO₂ emissions in industry i , $COAL_i$ and O_G_i are

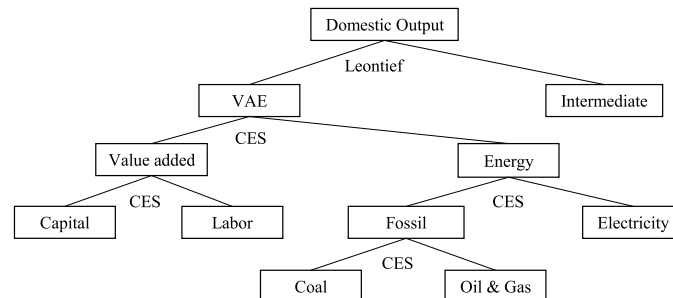


Fig. 1. The framework of production block.

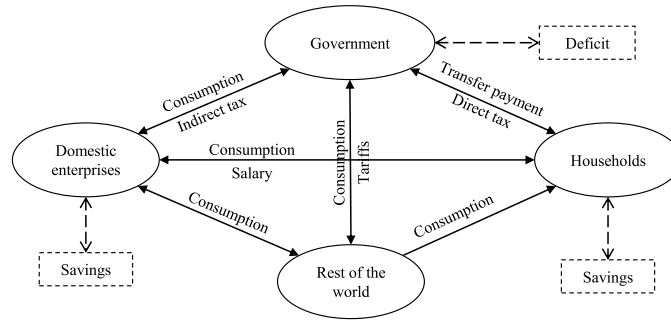


Fig. 2. The framework of income and expenditure block.

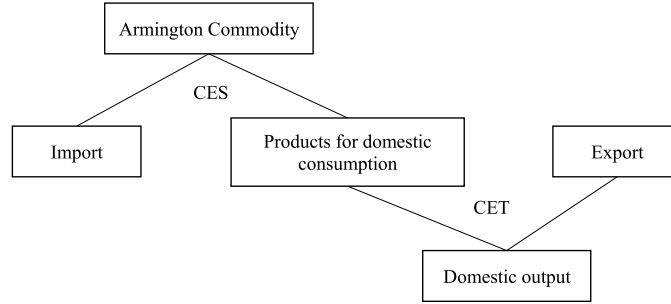


Fig. 3. the framework of trade block.

consumption of coal, and oil & gas in industry i in million tons of coal equivalent, γ^{coal} and γ^{o-g} are the coefficient of CO₂ emissions, PLC_{en} policy cost of the enterprises caused by a carbon tax in industry en (before offset), p^c is the carbon tax rate.

The CEEEA model assumes that the ETS market is completely competitive. Therefore, the ETS auction price is equal to the trading price of carbon emission allowances at equilibrium. If an enterprise's carbon emission is greater than its carbon emission allowances, the excess is fined by the government. This paper assumes that the revenue from the auction of carbon allowance and ETS fines are distributed between the government and residents. The free allowance rate benchmark is 0.9, which is based on the experience of seven ETS pilot cities and the phase I of EU-ETS. This free allowance rate means that 90% of carbon emission allowances are granted by the government to enterprises for free. Only 10% of CO₂ emission allowances can be auctioned and traded. The carbon trading block can be described by the following equations.

$$\begin{aligned} PLC_{ei} &= p^t (CA_{ei} - FA_{ei}) + p^f (EM_{ei} - CA_{ei}), EM_{ei} \geq CA_{ei} \\ PLC_{ei} &= p^t (CA_{ei} - FA_{ei}), EM_{ei} < CA_{ei} \end{aligned} \quad (3)$$

$$far = \sum_{ei} FA_{ei} / \sum_{ei} CA_{ei} \quad (4)$$

Where PLC_{ei} is policy cost caused by ETS in ETS coverage industry ei , p^t and p^f denote carbon allowance and free allowance in industry ei , far is free allowance rate.

On December 18, 2017, the national carbon trading market was officially launched in China, and only the power industry was allowed to participate in the carbon trading market.¹ The main reason is for easy Monitoring, Reporting, and Verification (MRV) in some provinces and industries. Due to the data quality, the plan of coverage industry is different from the plan put forward by the National Development and Reform Commission in China (NDRC) in January 2016. The 2016 NDRC

plan is that the first stage will cover eight major emission enterprises in petrochemical, chemical, building materials, steel, nonferrous metals, paper, power, and aviation industries.² According to the 2017 NDRC plan, the national carbon trading market is meant to cover the above eight key emission enterprises in the future. Therefore, the carbon trading coverage industry assumed in this paper will be based on the plan proposed by the national development and Reform Commission in January 2016. The detailed coverage assumption can be seen in Section 2.2.

In this paper, we assume that the free emission allowance of ETS coverage industry ei in the current year is determined by the output of this year, the output of last year, the carbon emission of last year and the annual decline coefficient, as shown in the Eq. (5).

$$FA_{ei,t} = Z_{ei,t} \times \frac{EM_{ei,t-1}}{Z_{ei,t-1}} \times (1 - \omega) \times far \quad (5)$$

Where $FA_{ei,t}$ is the free allowance of industry ei in current year t , $Z_{ei,t}$ and $Z_{ei,t-1}$ are the domestic output of industry ei in current year t and last year $t-1$, $EM_{ei,t-1}$ is the CO₂ emission of industry ei in current year t , ω is the annual decline coefficient. According to 6 ETS pilots, we set ω as 1%.

2.1.5. Macro closure and market clearing block

For the macro closure, this paper mainly considers three balance rules: government budget balance, foreign trade balance, and investment savings balance. These three principles are introduced in the income and expenditure block. CEEEA model assumes that all savings are converted into investment. The total investment equals endogenous total savings. Therefore, this model is savings-driven. Market clearing condition is used to express the balance of supply and demand of all markets in the model. This paper sets up two conditions for market-clearing, one is the clearing of commodity market: all Armington composite commodities are used for residents and government

¹ National Development and Reform Commission in China. the national carbon emission trading market construction plan (power generation industry), 2017, Available online: http://qhs.ndrc.gov.cn/qjzjz/201712/t20171220_871258.html

² National Development and Reform Commission in China. Do a good job in launching the national carbon emission trading market, 2016, Available online: http://www.ndrc.gov.cn/gzdt/201601/t20160122_772150.html

Table 1.
Sector classification and the coverage design in the carbon tax and emission trading.

Sectors	Abbreviation	ETS Coverage industry	CT Coverage industry
Agriculture, forestry, animal husbandry and fishery	AGR	–	–
Coal production	COL	–	yes
Oil and gas production	O_G	–	yes
Paper industry	PAP	yes	–
Cement industry	CMT	yes	–
Chemical fertilizer	FER	–	–
chemical industry	CMC	yes	–
Iron and steel industry	STL	yes	–
Equipment manufacturing industry	EQU	–	–
Power industry	ELC	yes	yes
construction industry	CST	yes	–
Transportation industry	TRA	yes	–
Other industries	OTH	–	–
Service industry	SER	–	–

consumption, savings, and intermediate input. Next is the clearing of the factor market: the quantity of labor force and capital of rural residents and urban residents is equal to the endowment of the labor force and capital of the whole society.

2.2. Social accounting matrix

Social Accounting Matrix (SAM) is the basic data of a CGE model. Most of the data are from the input-output table (China Input-Output Association, 2017). Some of the energy and emission data are from the China Statistical Yearbook (National Bureau of Statistics, 2015). There are two points worth noting. The first point is that the price discussed in this paper is the constant price in 2012, without considering the changes in inflation and consumer price index. The second point is that this paper only discusses the carbon dioxide emissions using fossil fuels in social activities, regardless of the carbon dioxide emissions generated by the decomposition of microorganisms and the respiration of animals and plants. For the convenience of analysis, this paper re-classifies the sectors in the social accounting matrix (see Table 1. As for the coverage industry design please refer to Section 2.1.4).

2.3. Dynamic method

CEEEA model considers population growth, capital accumulation, and technological progress as the dynamic component of the model using the recursive method. The capital endowment is determined by the current capital stock, depreciation rate, and investment. The labor endowment is exogenous, and the data comes from the *national population development plan (2016–2030)* (The Central People's Government of the People's Republic of China, 2017). The Automatic Energy Efficiency Improvement (AEEI) is considered as technological progress in the modeling: according to the *medium and long term plan for energy conservation and emission reduction* (National Development and Reform Commission (NDRC), 2005), the energy consumption level of China's energy-intensive industries will reach the world's advanced level in 2020. AEEI setting can be found in Table 2.

3. Scenario design

The scenario design of this paper simulates the possible carbon trading mechanism and carbon tax mechanism in China. Three scenarios are designed: business as usual scenario (BAU scenario), the scenario with emission trading scheme (ETS scenario) and the scenario with a carbon tax (CT scenario). The ETS scenario is based on the EU-ETS phase I and 7 ETS pilots in China (refer to Section 2.1.4); the CT

Table 2.
AEEI in each sector^a.

Sector	AEEI
AGR	0.025
COL	0.006
O_G	0.006
PAP	0.015
CMT	0.015
FER	0.02
CMC	0.015
STL	0.025
EQU	0.03
ELC	0.025
CST	0.006
TRA	0.033
OTH	0.016
SER	0.023

^a AEEI will be halved in 2020.

scenario is based on the assumption that energy producers are taxed in the carbon tax system and not all the consumers. Other settings are in line with Denmark and the Netherlands cases. The main purpose of this paper is to compare and analyze the emission reduction effect of the carbon tax and carbon trading under the condition that the economic impact is the same. Therefore, based on the ETS scenario, this paper changes the carbon tax rate in the CT scenario, so that the GDP of the ETS scenario and the CT scenario in 2030 are equal. It is shown that when the carbon tax rate is 55.5 CNY / ton, the GDP of the ETS scenario and that of the CT scenario in 2030 are equal. Finally, the carbon tax rate of CT scenario in this paper is set as 55.5 CNY / ton, which is in line with the carbon tax rate recommended by the relevant research group of the Ministry of Finance in China,³ as well as previous study Shi et al. (2019). The design of these scenarios is described in Table 3. We assume that CT and ETS are implemented in 2017.

The CGE model is a tool for scenario analysis. Presenting the logic and the mechanism is one of the advantages of this model. What we want to present is the impact mechanism in CT and ETS so that the choice of a starting point, such as 2030, is not extremely important. we choose 2030 as the starting point.

4. Results and discussions

4.1. Economic impact

4.1.1. GDP

The rate of change in GDP under the ETS, and CT scenarios relative to the BAU scenario is illustrated in Fig. 4. The negative impact of the CT scenario on the GDP reached a maximum of 0.120% in 2027. After that, the negative impact gradually decreased. The impact on GDP by 2030 is 0.116%. The negative impact of the ETS scenario on GDP reached a maximum of 0.127% in 2023. After that, the negative impact began to decrease slowly. The negative impact is 0.116% in 2030. It is not a coincidence that the impact of the carbon tax and carbon trading on GDP will reach the same level in 2030. However, from the perspective of the impact trend, we can draw the following conclusions: in the short term, the negative impact of a carbon tax on GDP is relatively smaller than that of ETS. However, in the long term, the negative impact of carbon trading on China's economy will be smaller than that of the carbon tax on China's economy. Moreover, whether it is a carbon tax or carbon trading, the negative impact on China's economy will not be greater than 0.2% in the context of emission reduction strategies. This may be an affordable cost for China.

³ Beijing business daily (2016): the restart button of carbon taxes will be pressed. Available online: <http://www.bbtnews.com.cn/2016/0929/162666.shtml>

Table 3.
Scenario design in this paper.

Scenario design	ETS scenario	CT scenario
Coverage industries	Papermaking, cement, chemical industry, steel, electricity, construction, transportation	Coal, oil & gas, electricity
Carbon tax rate	–	55.5 CNY/ton
Free allowance rate	0.9	–

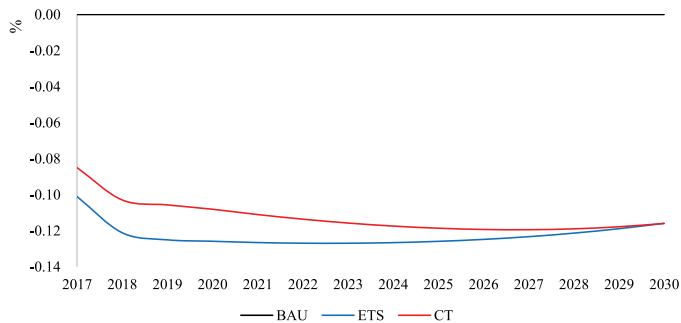


Fig. 4. GDP change rate in ETS and CT scenarios compared with the BAU scenario.

4.1.2. Sectoral output

Fig. 5 illustrates the change rates of sectoral output in 2030 for the CT and ETS scenarios compared with the BAU scenario. Carbon trading and carbon tax have great impacts on the output of the energy production industry, as well as the energy conversion industry. Coal production will be reduced by 5.06–5.18%, the oil and natural gas production is reduced by 2.74–2.81%, and power generation is reduced by 4.39–4.45%, while the reduction of other industries is relatively low. The output of other industries is reduced by less than 1%, except for 1–2% in cement, steel, equipment manufacturing and steel industry in the ETS scenario. The carbon tax policy has a positive impact on the agricultural and chemical industries. The output of these two industries increased by 0.18% and 0.04%, respectively. Carbon trading has a certain inhibitory effect on the output of the whole industry.

From these results, we can draw the following conclusions: 1) carbon tax and carbon trading can effectively reduce the output of the energy industry, hence, reduce the energy consumption of the society. 2) In addition to certain degrees of inhibition on the energy production industry, carbon trading also has an inhibition effect on the output of other industries, such as cement, steel, and equipment manufacturing industries. 3) According to the comparison of industrial output changes of coal, oil and natural gas, the ability of carbon tax to reduce energy

production is slightly higher than that of carbon trading under the same economic impact. 4) Under the carbon tax scenario, the decline in the service industry is significantly greater than the carbon trading scenario (the reasons for this phenomenon are explained in Section 4.1.3).

These findings suggest that when the carbon tax is levied on the energy industry, the energy industry will directly transfer the cost to all energy-using industries through the price mechanism of the commodity market. Therefore, in addition to the energy production industry, other industries are also subjected to different degrees of cost constraints. The degree of the constraint depends on the strength of the industry's energy demand. The mechanism of carbon trading is different. This is because of the pressure on increased energy-use cost. The coverage industries would increase their products' prices and reduce their output, which indirectly affects the price and output of other industries. Therefore, the reduction degree of coverage industries' output in the ETS scenario is generally higher than that in the CT scenario, while the reduction of the energy production industry is just the opposite. In other words, the carbon tax is a way of mitigation by increasing supply cost while carbon trading is a way of increasing demand cost.

4.1.3. Commodity price

Changes in commodity price in 2030, compared with the BAU scenario, is illustrated in Fig. 6. In the carbon tax scenario, the enterprises whose commodity prices are most affected are the power industry, with the price of electricity rising by 3.15%, followed by the price of coal, oil and gas industries. This is because the carbon tax is mainly levied on the energy supply side. It will directly push up the price of the energy production industry. Similarly, commodity prices in other industries will rise. Under the scenario of carbon trading, enterprises with higher prices are the industries covered by carbon trading, such as paper, cement, steel, electricity, etc. In general, the impact of a carbon tax on electricity commodity prices is greater than that of carbon trading while the impact on other commodity prices is less than that of carbon trading. However, since services require large electricity demand, the cost pressure on services under the carbon tax scenario is relatively higher than that of the carbon trading scenario. Therefore, the reduction of service output in the carbon tax scenario is

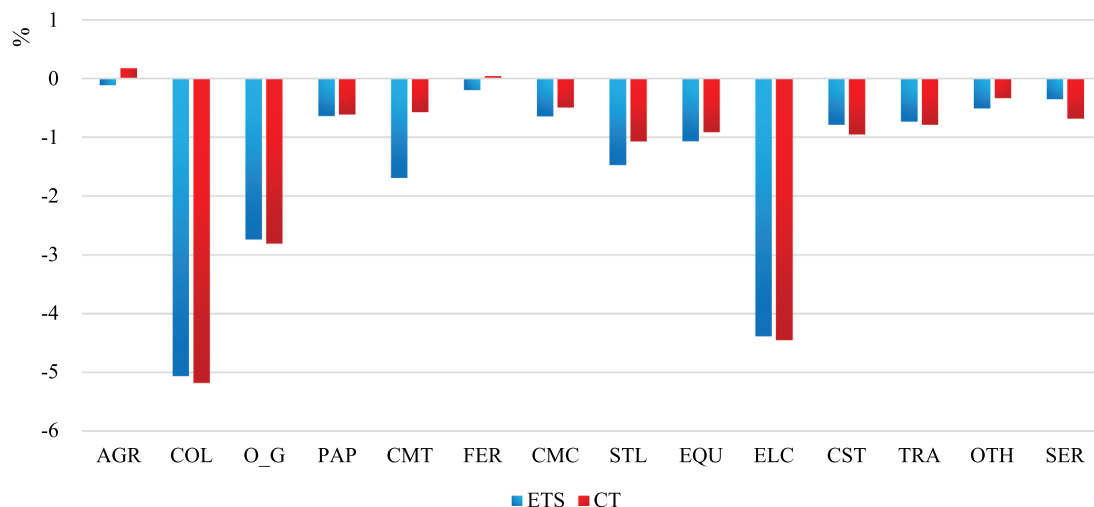


Fig. 5. Changes of sectoral output in 2030 compared with the BAU scenario.

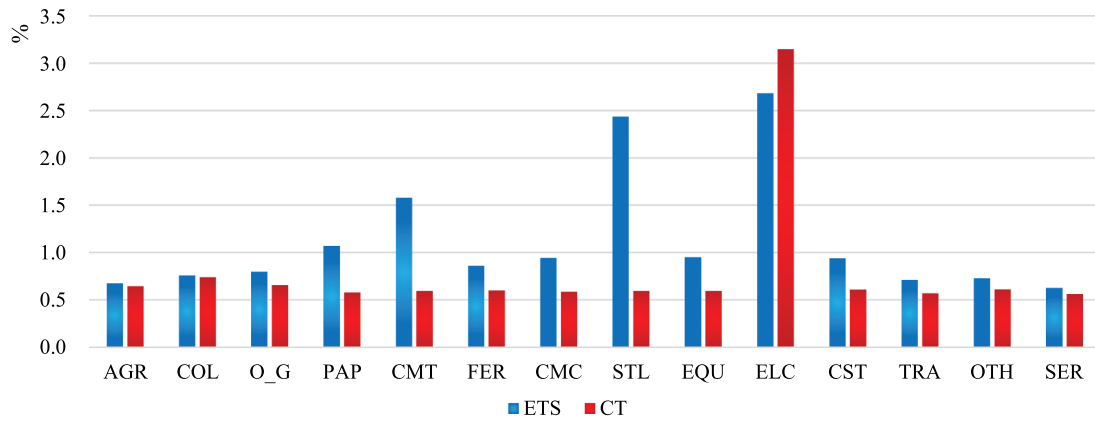


Fig. 6. Changes in commodity price in 2030 compared with the BAU scenario.

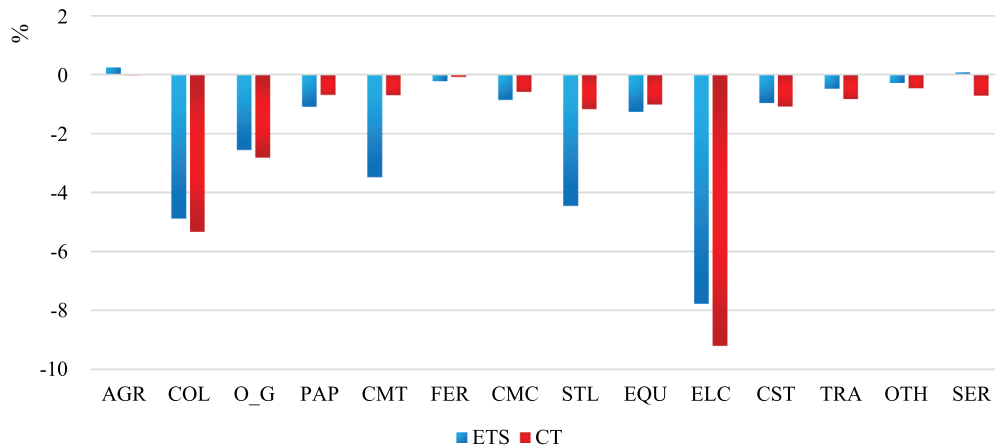


Fig. 7. Export changes in 2030 compared with the BAU scenario.

slightly higher than that in the carbon trading scenario, which also explains the different output performance between the two scenarios in Section 4.1.2.

4.1.4. International trade

Figs. 7 and 8 show the changes in export and import in 2030 for both ETS and CT scenarios. Recall from Section 4.1.2, “carbon tax is a way of emission mitigation by increasing supply cost, and carbon trading is by increasing demand cost”. The results of international trade also provide the same evidence. A carbon tax will increase the domestic production cost of energy products. Therefore, import increases and export decreases in the CT scenario than in the ETS scenario. Secondly, the export of energy-intensive industries will fall, especially for those covered in an emission trading scheme, but for the carbon tax condition, the decline in exports for these industries is not large. This is because the commodity prices of these industries are higher under the context of ETS.

4.2. Energy and environmental impact

4.2.1. Fossil energy consumption

Fig. 9 shows changes in primary consumption in 2030 compared with the BAU scenario. Carbon trading and carbon tax have a certain degree of inhibition on the energy consumption of various industries. Energy consumption by the energy production industry has the most reduction. This is closely related to the reduction of its output. The energy consumption by the energy production industry declines by 5.09–7.39%. The energy consumption by the coal industry declines the most, followed by the power industry, and then, the oil and gas

industry. By comparing Figs. 5 and 9, it is seen that the energy intensity of all industries is reduced (the decline in sectoral output is smaller than that of energy consumption). The energy consumption of cement, chemical industry, steel and other industries covered by carbon trading decreases more in the carbon trading scenario while the energy consumption of the energy production industry, transportation, and service industries decreases more in the carbon tax scenario. This is caused by the choice of the ETS coverage industry and the energy intensity of the industries.

4.2.2. CO₂ emission reduction

Table 4 shows the CO₂ emission reduction in the ETS and CT scenarios during 2017–2030. For either the carbon trading or carbon tax scenario, the emission reduction capacity is significant. In 2017, the annual emission reduction of carbon trading is 304 million tons, while the carbon tax is 254 million tons. Over time, the annual emission reduction effect of the two emission reduction policies increases. but the increase in the carbon tax is significantly greater than that of carbon trading. In 2029, the emission reduction of the two policies is the same. In 2030, the annual emission reduction in carbon tax reached 596 million tons, more than 589 million tons of carbon trade. From 2017 to 2030, the accumulated emission reduction for carbon trade and carbon tax policies reached 6440 and 6006 million tons respectively, which is about 91.95% and 85.75% of China's annual carbon emissions in 2010.

In the short term, the emission reduction effect of carbon trading is better than that of the carbon tax policy but this advantage will gradually reduce over time. In the long term, carbon tax's emission reduction effect is greater than carbon trading. However, from Fig. 4, it is clear that in the long run, the impact of the carbon tax on the economy

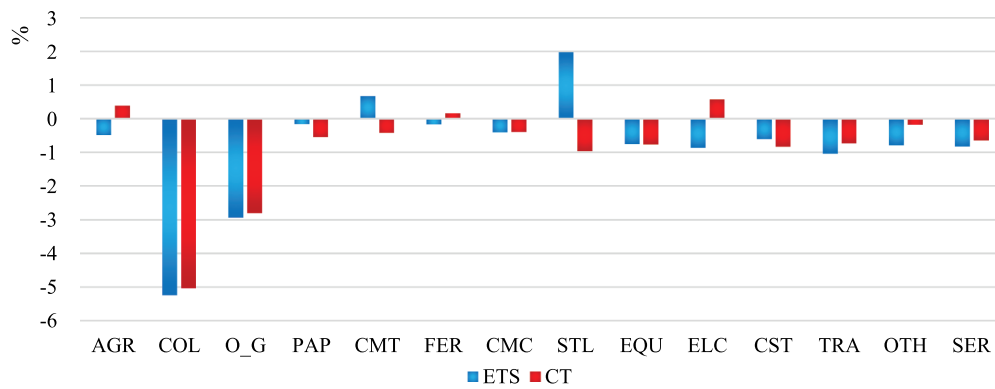


Fig. 8. Import changes in 2030 compared with the BAU scenario.

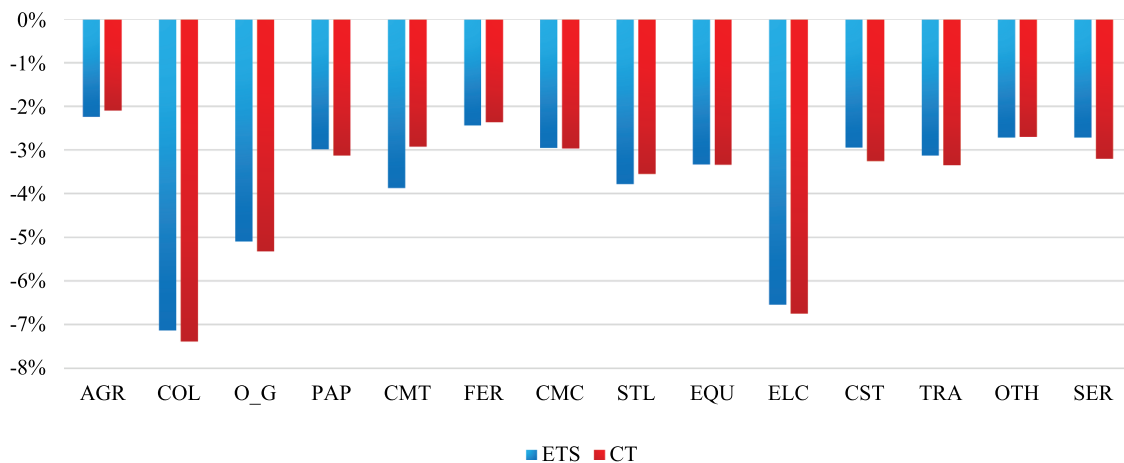


Fig. 9. Changes in primary consumption in 2030 compared with the BAU scenario.

Table 4.

CO₂ emission reduction in all scenarios compared with the BAU scenario (unit: million tons of CO₂).

Year	ETS	CT
2017	304	254
2018	362	313
2019	385	330
2020	401	349
2021	417	369
2022	434	390
2023	450	412
2024	468	435
2025	486	459
2026	505	484
2027	525	510
2028	546	538
2029	567	566
2030	589	596
Total	6440	6006

may be greater than carbon trading. Thus, so it is of great significance to compare the impact of the two emission reduction strategies on carbon emission intensity.

4.2.3. Carbon emission intensity

Carbon emission intensity is one of the indicators to measure carbon emission efficiency. Its calculation rule is the CO₂ emissions divided by an economic indicator such as the GDP. The intensity of carbon dioxide emissions in all scenarios from 2017 to 2030 is shown in Fig. 10. Under the BAU scenario, the carbon emission intensity decreased from 167.1 tons / million CNY in 2017 to 145.4 tons / million CNY in 2030. Carbon

trading and tax can significantly reduce carbon emission intensity. The carbon intensity of the ETS scenario and CT scenario gradually decreased from 161.8 and 162.7 tons / million CNY in 2017 to 138.8 and 138.7 tons / million CNY in 2030. Over time, the carbon intensity gap between the BAU scenario and the ETS or CT scenarios gradually increased. This proves the effectiveness of these two emission reduction policies. By comparing the ETS scenario and the CT scenario, we can observe that within 2017–2028, the carbon emission intensity of carbon trading scenario is less than that of the carbon tax, but after 2029, the carbon tax has a better performance via carbon intensity. To some extent, this shows that in the long run, the carbon tax has a stronger positive effect on optimizing resource allocation and reducing carbon emission intensity. The main reason could be the flaws in the carbon market. Although the ETS can increase the enterprise's emission cost and promote energy conservation and emission reduction in the coverage industries, it can only affect the behaviors of the industries it covers. The carbon tax mechanism increases the cost of energy supply through taxes which leads to the rise in the product price of the energy production industry and reduces energy-related carbon emissions in the whole society through relative price changes. The industry of carbon tax may cover a wider range through the commodity market, and it is easier to achieve a rational allocation of resources, to improve energy efficiency and reduce carbon intensity.

4.2.4. Reduction cost

Table 5 shows the reduction cost within 2017–2030 in the ETS and CT scenarios. The carbon emission reduction cost is a measure of GDP loss per unit of CO₂ emission. It's calculated as GDP loss divided by carbon dioxide emission reduction. In the ETS scenario, the emission reduction cost was 186.28 CNY / ton in 2017 and gradually increase to 195.53 CNY / ton in 2020. Thereafter, it decreases to 171.03 CNY / ton

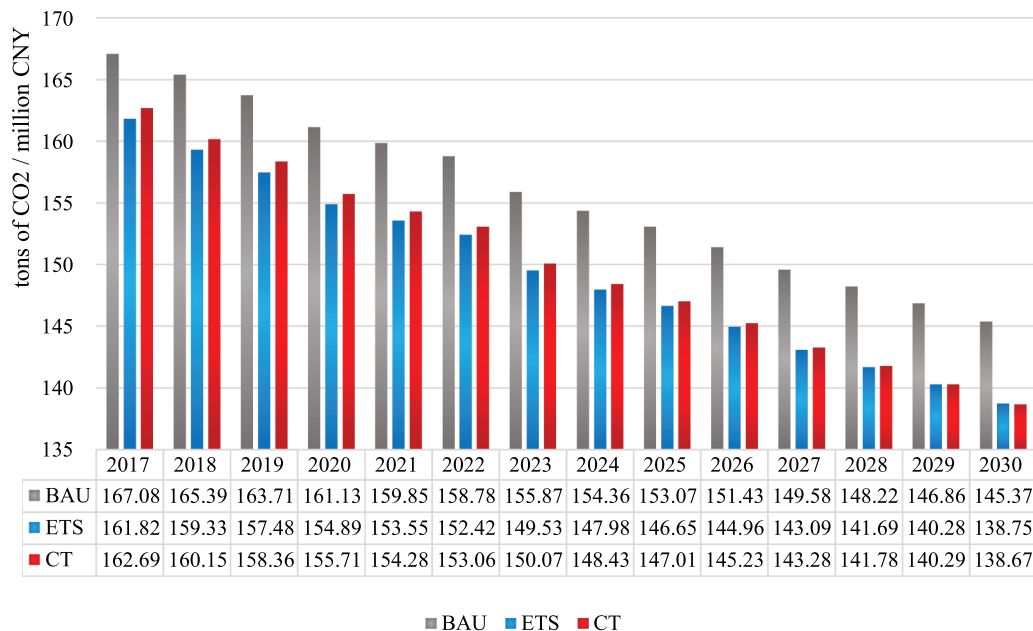


Fig. 10. Carbon emission intensity in all scenarios during 2017–2030.

Table 5.

Cost of emission reduction in ETS and CT scenarios (unit: CNY / ton).

Year	ETS	CT
2017	186.28	187.57
2018	193.66	190.48
2019	194.69	191.68
2020	195.53	193.30
2021	195.09	193.57
2022	193.90	192.94
2023	194.51	193.91
2024	192.88	192.49
2025	190.44	190.11
2026	187.94	187.51
2027	185.08	184.43
2028	181.01	180.01
2029	176.28	174.82
2030	171.03	168.99
Average	188.45	187.27

in 2030. In CT scenario, the cost of emission reduction in 2017 is 187.57 CNY / ton, and gradually rise to 193.57 CNY / ton in 2021 and reduces to 168.99 CNY / ton in 2030. Generally, the emission reduction cost of the carbon tax and carbon trading shows a similar trend of rising and declining afterwards. With time, the reduction amount of carbon tax emission reduction cost exceeds that of carbon trading. From 2017 to 2030, the average carbon emission reduction cost of the carbon tax mechanism is also smaller than that of carbon trading. Hence, the carbon tax has some advantages over carbon trading.

5. Conclusion and policy implications

5.1. Conclusion

By building a CEEEA model with the carbon tax and carbon trading scenarios, we analyze the impacts of emission trading and a carbon tax on energy production, economy, and environment at a given level of GDP in 2030. We provide analysis of the carbon tax and carbon trading mechanisms on the sectoral output, commodity price, energy consumption, carbon dioxide emission reduction, carbon emission intensity and emission reduction cost.

The results show that carbon tax is a way of mitigation by increasing supply cost while carbon trading is by increasing demand cost. To be specific, the carbon tax on energy enterprises will directly increase the cost of the industries, then the domestic supply cost of energy products increases while the emission trading scheme impacts differently. It reduces emissions by increasing the cost of energy users, especially energy-intensive industries. This implies that the demand for these industries for energy leads to additional costs. These mechanisms have different influence path. A carbon tax directly reduces energy production and indirectly reduces carbon dioxide emissions caused by energy consumption after international and domestic trade. Carbon trading directly reduces the carbon emissions of covered industries but has a limited impact on other uncovered industries. The carbon tax leads to an additional cost for energy users and that additional cost is accumulated through the industrial chain, like pricing on carbon footprint. Therefore, levying carbon tax on energy production enterprises may be the most effective way to reduce carbon emissions theoretically.

Specific results in this paper are shown as follows:

In the short term, the negative impacts of a carbon tax on China's GDP and the positive impacts of carbon dioxide emission reduction are relatively low. In the long term, the impact of a carbon tax on China's GDP and emission reduction will gradually be greater than the impact of carbon trading in China. We also found that carbon trading has a certain inhibitory effect on the energy production industry, as well as the output of other covered industries, such as cement, steel, and equipment manufacturing industry. Under the same economic impact, the ability of carbon tax to reduce energy consumption is slightly higher than that of carbon trading. The negative impact of a carbon tax on the service industry's output is greater than that of carbon trading (see Sections 4.1.2 and 4.1.3). Since the carbon tax is mainly levied on the energy supply side, a carbon tax will directly promote the price of the energy production industry. At the same time, due to the rising cost, the commodity price of other industries will rise slightly. Under the scenario of carbon trading, enterprises with higher prices are industries covered by carbon trading, such as papermaking, cement, steel, power industry, etc.

From 2017 to 2030, the total emission reductions of carbon trading and carbon tax policies reached 6.44 billion tons and 6.01 billion tons respectively, accounting for 91.95% and 85.75% of China's annual carbon emissions in 2010. In the long run, the emission reduction capacity of a carbon tax is slightly greater than that of a carbon trading

mechanism. In the early stages of the policy implementation, the carbon dioxide emission intensity of the carbon tax is greater than that of the carbon trading, but after 2029, this relationship is reversed, that is to say, the carbon tax will have a longer-term advantage over the carbon trading in terms of emission reduction efficiency. In general, the cost of carbon tax reduction is lower than that of carbon trading.

Under the same GDP impact, both carbon trading and carbon tax have strong emission reduction capacity, but the relative emission reduction efficiency of a carbon tax is higher than that of carbon trading. This advantage increases with time. This is different from most previous studies. Many studies believe that the carbon trading mechanism is a market-based mechanism, so it has more advantages than a simple and crude carbon tax. However, this paper argues that both mechanisms are market-based strategies for emission reduction. Implementing a carbon tax can also change the production behavior and market behavior via price changes, hence, the carbon tax is another carbon pricing strategy. The way to achieve emission reduction via the carbon trading mechanism is to build a new cap and trade market. Through the price mechanism of carbon trading price, carbon emissions are reallocated in coverage industries for profit maximization. In theory, it is the best way to reduce emissions among the coverages. The approach for a carbon tax is to “borrow” the previously mature commodity market. Through the commodity price mechanism, all energy users can redistribute and integrate internal resources to achieve the goal of emission reduction. The increase in price caused by a carbon tax can be understood as the implied carbon trading price.

Whether the emission reduction strategies are carbon trading mechanism or carbon tax mechanism, they all have advantages and certain design problems.

- 1) The carbon market is a financial market in nature, with arbitrage space and relatively small resistance. There are also several problems in carbon trading: carbon trading needs to build a unified trading platform, which will cost a significant amount of social resources (such as monitoring, reporting, and verification). Although the carbon trading mechanism is based on a new market with cap and trade (so that enterprises can trade freely in theory and achieve the optimization of carbon emission allowances), there are many problems in the ETS mechanism design, such as how to set the coverage, how to measure the cap, how to avoid rent-seeking behavior, and so on. Especially, there are obvious risks in the carbon trading market i.e. high carbon prices will lead to high costs in the enterprises. The enterprises are more willing to take risks to evade

regulation, which will eventually lead to the rise of regulatory costs and waste of social resources.

- 2) The main problem of a carbon tax is about “who should be in the taxation system and the measurement of tax rate?” The tax industry has got a relatively unified understanding across the economies of the world, but the carbon tax rate should adapt to local conditions. In terms of the mechanism setting, the design and operation of the carbon tax are easier and simpler. By collecting taxes on the production (or consumption) of primary energy, the price of energy can be directly (or indirectly) affected. This method may be crude, as there is no unified market, the carbon tax rate is difficult to measure, as well as the effects. More importantly, the trading of energy commodities after the carbon tax is imposed is a process of resource reallocation due to price changes. Therefore, this paper shows that the carbon tax is a more effective emission reduction mechanism. The regulation is relatively simple, and it is more difficult to evade regulation compared with emission trading because the regulatory scope is smaller.

5.2. Policy implications

This paper shows that in the long term, a carbon tax may have a better emission mitigation performance. Maybe it's because the coverage of emission trading is not wide enough so that the theoretical effectiveness is not reached. Higher coverage usually means higher cost of MRV (Monitoring, Reporting, and Verification), especially in developing countries. Lower coverage range may curb the economic structure if the carbon pricing is too high. The carbon tax can be understood as a recessive carbon trading market with full coverage, which affects energy exports. As long as carbon pricing is reasonable (though it is difficult), its long-term efficiency should be higher. Those factors of ETS may not have a better effect on emission reduction efficiency than CT.

This paper suggests that China can levy a carbon tax in the form of an energy tax on energy production enterprises. This is because carbon tax has better emission reduction efficiency, lower emission reduction cost in the long term based on the same impact on GDP. However, based on the current national carbon trading market, this paper proposes the following three implications based on the current national ETS conditions and research findings, which are abstracted in Fig. 11.

- 1) At present, China's carbon trading market only covers the power industry. This paper proposes to cover other energy production industries such as coal, oil and natural gas in the future. In this way,

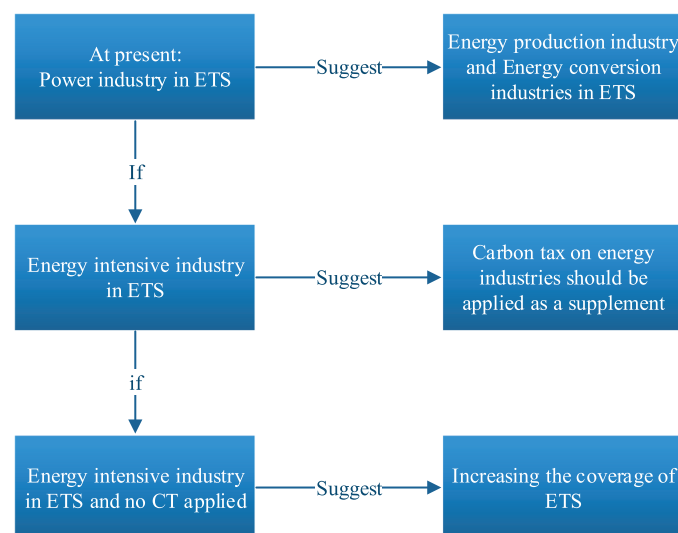


Fig. 11. The policy implications in this paper.

we can simulate the way of levying carbon tax on the energy industry to push up the energy price to a certain extent, and then it can promote the whole industry to achieve the purpose of emission reduction at a lower cost through the relative price effect.

- 2) If other energy-intensive industries are to be covered according to the 2016 NDRC's plan, this paper suggests that the carbon tax is levied on energy enterprises as a supplement. At the same time, to reduce the burden of the energy-intensive enterprises, the cap of carbon allowance can be increased slightly or the annual decline coefficient of allowance can be reduced. This can also promote the improvement of emission reduction efficiency and reduce the cost of emission reduction to some extent.
- 3) If other energy-intensive industries are to be covered according to the 2016 NDRC's plan, and the carbon tax is not implemented as a supplementary emission reduction strategy. This paper suggests that we can improve the efficiency of emission reduction and reduce the cost of emission reduction by improving the coverage of the carbon

trading market. The specific implementation policies include reducing the threshold of coverage and increasing industry coverage, etc.

Conflicts of Interest

We declare no conflict of interest.

CRediT authorship contribution statement

Zhijie Jia: Methodology, Software, Data curation, Writing - original draft. **Boqiang Lin:** Conceptualization, Methodology, Software, Data curation, Writing - original draft.

Acknowledgments

The paper is supported by Report Series from Ministry of Education of China (No.10JBG013).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.techfore.2020.120187](https://doi.org/10.1016/j.techfore.2020.120187).

Appendix A CEEEA/CGE model in this paper

A.1 Production block

$$NOE_i = \alpha_i^{noe} \left[\delta_i^{noe} COAL_i^{\rho_i^{noe}} + (1 - \delta_i^{noe}) NOS_i^{\rho_i^{noe}} \right]^{1/\rho_i^{noe}} \quad (A.1)$$

$$\frac{PCOAL_i}{PNOS_i} = \frac{\delta_i^{noe}}{1 - \delta_i^{noe}} \left(\frac{NOS_i}{COAL_i} \right)^{(1-\rho_i^{noe})} \quad (A.2)$$

$$NOE_i PNOE_i = COAL_i PCOAL_i + NOS_i PNOS_i \quad (A.3)$$

$$ENE_i = \alpha_i^{ene} \left[\delta_i^{ene} ELE_i^{\rho_i^{ene}} + (1 - \delta_i^{ene}) NOE_i^{\rho_i^{ene}} \right]^{1/\rho_i^{ene}} \quad (A.4)$$

$$\frac{PELE_i}{PNOE_i} = \frac{\delta_i^{ene}}{1 - \delta_i^{ene}} \left(\frac{NOE_i}{ELE_i} \right)^{(1-\rho_i^{ene})} \quad (A.5)$$

$$ENE_i PENE_i = ELE_i PELE_i + NOE_i PNOE_i \quad (A.6)$$

$$VA_i = \alpha_i^{va} \left[\delta_i^{va} LAB_i^{\rho_i^{va}} + (1 - \delta_i^{va}) CAP_i^{\rho_i^{va}} \right]^{1/\rho_i^{va}} \quad (A.7)$$

$$\frac{PLAB_i}{PCAP_i} = \frac{\delta_i^{va}}{1 - \delta_i^{va}} \left(\frac{CAP_i}{LAB_i} \right)^{(1-\rho_i^{va})} \quad (A.8)$$

$$VA_i PVA_i = LAB_i PLAB_i + CAP_i PCAP_i \quad (A.9)$$

$$VAE_i = \alpha_i^{vae} \left[\delta_i^{vae} VA_i^{\rho_i^{vae}} + (1 - \delta_i^{vae}) ENE_i^{\rho_i^{vae}} \right]^{1/\rho_i^{vae}} \quad (A.10)$$

$$\frac{PVA_i}{PENE_i} = \frac{\delta_i^{vae}}{1 - \delta_i^{vae}} \left(\frac{ENE_i}{VA_i} \right)^{(1-\rho_i^{vae})} \quad (A.11)$$

$$VAE_i PVAE_i = ENE_i PENE_i + VA_i PVA_i \quad (A.12)$$

$$INT_{i,j} = a_{i,j}^{INT} Z_j \quad (A.13)$$

$$FVAE_j = a_j^{VAE} Z_j \quad (A.14)$$

$$PZ_j = a_j^{vae} PVAE + \sum_i a_{i,j}^{INT} PQ_i \quad (A.15)$$

A.2 Income and expenditure block

$$SP_l = SS_l^p \sum_i \gamma_l^{lab} LAB_i \cdot PLAB_i + \gamma_l^{cap} CAP_i \cdot PCAP_i \quad (A.16)$$

$$SG = SS^g \left(\sum_l TD_l + \sum_i TZ_i + \sum_i TM_i \right) \quad (A.17)$$

$$XG_i = \frac{\mu_i}{PQ_i} \left(\sum_l TD_l + \sum_i TZ_i + \sum_i TM_i - SG \right) \quad (A.18)$$

$$XP_{i,l} = \frac{\beta_{i,l}^{xp}}{PQ_i} \left(\sum_i (\gamma_l^{lab} LAB_i \cdot PLAB_i + \gamma_l^{cap} CAP_i \cdot PCAP_i) - SP_l - TD_l \right) \quad (A.19)$$

$$TD_l = \tau_l^d \sum_i (\gamma_l^{lab} LAB_i \cdot PLAB_i + \gamma_l^{cap} CAP_i \cdot PCAP_i) \quad (A.20)$$

$$TZ_i = \tau^z PZ_i Z_i (1 - cor_i) + PLC_i \quad (A.21)$$

$$TM_i = \tau^m PM_i M_i \quad (A.22)$$

A.3 Trade block

$$PE_i = \varepsilon PWE_i \quad (A.23)$$

$$PM_i = \varepsilon PWM_i \quad (A.24)$$

$$\sum_i PWE_i E_i + SF = \sum_i PWM_i M_i \quad (A.25)$$

$$Q_i = \gamma_i (\delta m_i M_i^{\eta_i} + \delta d_i D_i^{\eta_i})^{1/\eta_i} \quad (A.26)$$

$$M_i = \left[\frac{\gamma_i^{\eta_i} \delta m_i PQ_i}{(1 + \tau_i^m) PM_i} \right]^{\frac{1}{1-\eta_i}} Q_i \quad (A.27)$$

$$D_i = \left[\frac{\gamma_i^{\eta_i} \delta d_i PQ_i}{PD_i} \right]^{\frac{1}{1-\eta_i}} Q_i \quad (A.28)$$

$$Z_i = \theta_i (\xi e_i E_i^{\phi_i} + \xi d_i D_i^{\phi_i})^{\frac{1}{\phi_i}} \quad (A.29)$$

$$E_i = \left[\frac{\theta_i^{\phi_i} \xi e_i [(1 + \tau_i^z)(1 - cor_i) + PLC_i / PZ_i / Z_i] PZ_i}{PE_i} \right]^{\frac{1}{1-\phi_i}} Z_i \quad (A.30)$$

$$D_i = \left[\frac{\theta_i^{\phi_i} \xi d_i [(1 + \tau_i^z)(1 - cor_i) + PLC_i / PZ_i / Z_i] PZ_i}{PD_i} \right]^{\frac{1}{1-\phi_i}} Z_i \quad (A.31)$$

A.4 Carbon tax and carbon trading block

$$EM_i = ENE_COAL_i \times \gamma^{coal} + ENE_O_G_i \times \gamma^{o-g} \quad (A.32)$$

$$COAL_i = \chi_i^{coal} \times ENE_COAL_i \quad (A.33)$$

$$NOS_i = \chi_i^{nos} \times ENE_O_G_i \quad (A.34)$$

$$PLC_{en} = p^{ct} EM_{en} \text{ in CT scenario}$$

$$PLC_{ei} = p^t (CA_{ei} - FA_{ei}) + p^f (EM_{ei} - CA_{ei}), EM_{ei} \geq CA_{ei} \text{ in ETS scenario} \\ PLC_{ei} = p^t (CA_{ei} - FA_{ei}), EM_{ei} < CA_{ei} \quad (A.35)$$

$$far = \sum_{ei} FA_{ei} / \sum_{ei} CA_{ei} \text{ in ETS scenario} \quad (A.36)$$

A.5 Macro closure and market clearing block

$$XV_i = \frac{\lambda_i}{PQ_i} \left(\sum_l SP_l + SG + \varepsilon SF \right) \quad (\text{A.37})$$

$$Q_i = \sum_l XP_{i,l} + XG_i + XV_i + \sum_j X_{i,j} \quad (\text{A.38})$$

$$\sum_i LAB_i = \sum_l FF_l^{lab} \quad (\text{A.39})$$

$$\sum_i CAP_i = \sum_l FF_l^{cap} \quad (\text{A.40})$$

References

- Andersson, J.J., 2019. Carbon taxes and CO2 emissions: sweden as a case study. *Am. Econ. J. Econ. Policy* 11, 1–30. <https://doi.org/10.1257/pol.20170144>.
- Barragán-Beaud, C., Pizarro-Alonso, A., Xylia, M., Syri, S., Silveira, S., 2018. Carbon tax or emissions trading? An analysis of economic and political feasibility of policy mechanisms for greenhouse gas emissions reduction in the Mexican power sector. *Energy Policy* 122, 287–299. <https://doi.org/10.1016/j.enpol.2018.07.010>.
- Berry, A., 2019. The distributional effects of a carbon tax and its impact on fuel poverty: a microsimulation study in the French context. *Energy Policy* 124, 81–94. <https://doi.org/10.1016/j.enpol.2018.09.021>.
- Bhat, A.A., Mishra, P.P., 2020. Evaluating the performance of carbon tax on green technology: evidence from India. *Environ. Sci. Pollut. Res.* 27, 2226–2237. <https://doi.org/10.1007/s11356-019-06666-x>.
- Chan, N.W., Morrow, J.W., 2019. Unintended consequences of cap-and-trade? Evidence from the regional greenhouse gas initiative. *Energy Econ.* 80, 411–422. <https://doi.org/10.1016/j.eneco.2019.01.007>.
- Chang, K., Zhang, C., Chang, H., 2016. Emissions reduction allocation and economic welfare estimation through interregional emissions trading in China: evidence from efficiency and equity. *Energy* 113, 1125–1135. <https://doi.org/10.1016/j.energy.2016.07.113>.
- Chao, H., Agusdinata, D.B., DeLaurentis, D.A., 2019. The potential impacts of emissions trading scheme and biofuel options to carbon emissions of U.S. airlines. *Energy Policy* 134, 110993. <https://doi.org/10.1016/j.enpol.2019.110993>.
- China Input-Output Association, 2017. 2012 Input-Output Table [WWW Document]. URL http://www.stats.gov.cn/zjtj/tjzdg/trccxh/zlxz/trccb/201701/t20170113_1453448.html (accessed 4.16.18).
- Craig, M.T., Zhai, H., Jaramillo, P., Klimm, K., 2017. Trade-offs in cost and emission reductions between flexible and normal carbon capture and sequestration under carbon dioxide emission constraints. *Int. J. Greenh. Gas Control* 66, 25–34. <https://doi.org/10.1016/j.ijggc.2017.09.003>.
- Dorsey-Palmeater, R., Niu, B., 2020. The effect of carbon taxation on cross-border competition and energy efficiency investments. *Energy Econ.* 85, 104602. <https://doi.org/10.1016/j.eneco.2019.104602>.
- Du, K., Li, P., Yan, Z., 2019. Do green technology innovations contribute to carbon dioxide emission reduction? Empirical evidence from patent data. *Technol. Forecast. Soc. Change* 146, 297–303. <https://doi.org/10.1016/j.techfore.2019.06.010>.
- Forero-Cantor, G., Ribal, J., Sanjuán, N., 2020. Levying carbon footprint taxes on animal-sourced foods. A case study in Spain. *J. Clean. Prod.* 243, 118668. <https://doi.org/10.1016/j.jclepro.2019.118668>.
- Fremstad, A., Paul, M., 2019. The impact of a carbon tax on inequality. *Ecol. Econ.* 163, 88–97. <https://doi.org/10.1016/j.ecolecon.2019.04.016>.
- Frey, M., 2017. Assessing the impact of a carbon tax in Ukraine. *Clim. Policy*. <https://doi.org/10.1080/14693062.2015.1096230>.
- Gonzalez, F., 2012. Distributional effects of carbon taxes: the case of Mexico. *Energy Econ* 34, 2102–2115. <https://doi.org/10.1016/j.eneco.2012.03.007>.
- Goulder, L.H., Hafstead, M.A.C., Kim, G., Long, X., 2019. Impacts of a carbon tax across US household income groups: what are the equity-efficiency trade-offs? *J. Public Econ* 175, 44–64. <https://doi.org/10.1016/j.jpubeco.2019.04.002>.
- Hájek, M., Zimmermannová, J., Helman, K., Rozenský, L., 2019. Analysis of carbon tax efficiency in energy industries of selected EU countries. *Energy Policy* 134, 110955. <https://doi.org/10.1016/j.enpol.2019.110955>.
- Higgins, A., Foliente, G., McNamara, C., 2011. Modelling intervention options to reduce GHG emissions in housing stock — A diffusion approach. *Technol. Forecast. Soc. Change* 78, 621–634. <https://doi.org/10.1016/j.techfore.2010.12.003>.
- Hu, X., Yang, Z., Sun, J., Zhang, Y., 2020. Carbon tax or cap-and-trade: which is more viable for Chinese remanufacturing industry? *J. Clean. Prod.* 243, 118606. <https://doi.org/10.1016/j.jclepro.2019.118606>.
- Jiang, Z., Shao, S., 2014. Distributional effects of a carbon tax on Chinese households: a case of Shanghai. *Energy Policy* 73, 269–277. <https://doi.org/10.1016/j.enpol.2014.06.005>.
- Kong Chyong, C., Guo, B., Newbery, D., 2020. The impact of a carbon tax on the CO2 emissions reduction of wind. *Energy J.* 41. <https://doi.org/10.5547/01956574.41.1.cchy>.
- Li, A., Peng, D., Wang, D., Yao, X., 2017. Comparing regional effects of climate policies to promote non-fossil fuels in China. *Energy* 141, 1998–2012. <https://doi.org/10.1016/j.energy.2017.11.108>.
- Lin, B., Jia, Z., 2019. What will China's carbon emission trading market affect with only electricity sector involvement? A CGE based study. *Energy Econ.* 78, 301–311. <https://doi.org/10.1016/j.eneco.2018.11.030>.
- Lin, B., Jia, Z., 2018. The energy, environmental and economic impacts of carbon tax rate and taxation industry: a CGE based study in China. *Energy* 159, 558–568. <https://doi.org/10.1016/j.energy.2018.06.167>.
- Lin, W., Chen, J., Zheng, Y., Dai, Y., 2019. Effects of the EU Emission Trading Scheme on the international competitiveness of pulp-and-paper industry. *For. Policy Econ.* 109, 102021. <https://doi.org/10.1016/j.forpol.2019.102021>.
- Liu, C., Ma, C., Xie, R., 2020. Structural, innovation and efficiency effects of environmental regulation: evidence from China's carbon emissions trading pilot. *Environ. Resour. Econ.* 75, 741–768. <https://doi.org/10.1007/s10640-020-00406-3>.
- Ma, N., Li, H., Wang, Y., Zhang, J., Li, Z., Arif, A., 2020. The short-term roles of sectors during a carbon tax on Chinese economy based on complex network: an in-process analysis. *J. Clean. Prod.* 251, 119560. <https://doi.org/10.1016/j.jclepro.2019.119560>.
- Mu, Y., Evans, S., Wang, C., Cai, W., 2018. How will sectoral coverage affect the efficiency of an emissions trading system? A CGE-based case study of China. *Appl. Energy* 227, 403–414. <https://doi.org/10.1016/j.apenergy.2017.08.072>.
- National Bureau of Statistics, 2015. China Statistical Yearbook (2014) [WWW Document]. URL <http://www.stats.gov.cn/tjsj/ndsj/> (accessed 2.3.18).
- National Development and Reform Commission (NDRC), 2005. Medium and long-term energy saving special planning [WWW Document]. URL http://xwzx.ndrc.gov.cn/xwfb/200506/t20050628_104993.html.
- Nong, D., 2020. Development of the electricity-environmental policy CGE model (GTAP-E-PowerS): a case of the carbon tax in South Africa. *Energy Policy* 140, 111375. <https://doi.org/10.1016/j.enpol.2020.111375>.
- Nong, D., Nguyen, T.H., Wang, C., Van Khuc, Q., 2020. The environmental and economic impact of the emissions trading scheme (ETS) in Vietnam. *Energy Policy* 140, 111362. <https://doi.org/10.1016/j.enpol.2020.111362>.
- Ntombela, S.M., Bohlmann, H.R., Kalaba, M.W., 2019. Greening the South Africa's Economy Could Benefit the Food Sector: evidence from a Carbon Tax Policy Assessment. *Environ. Resour. Econ.* 74, 891–910. <https://doi.org/10.1007/s10640-019-00352-9>.
- Olale, E., Yiridoe, E.K., Ochuodho, T.O., Lantz, V., 2019. The effect of carbon tax on farm income: evidence from a Canadian Province. *Environ. Resour. Econ.* 74, 605–623. <https://doi.org/10.1007/s10640-019-00337-8>.
- Pearce, D., 1991. The role of carbon taxes in adjusting to global warming. *Econ. J.* <https://doi.org/10.2307/2233865>.
- Peng, Z.-S., Zhang, Y.-L., Shi, G.-M., Chen, X.-H., 2019. Cost and effectiveness of emissions trading considering exchange rates based on an agent-based model analysis. *J. Clean. Prod.* 219, 75–85. <https://doi.org/10.1016/j.jclepro.2019.01.303>.
- Qian, W., Suryani, A.W., Xing, K., 2020. Does carbon performance matter to market returns during climate policy changes? Evidence from Australia. *J. Clean. Prod.* 259, 121040. <https://doi.org/10.1016/j.jclepro.2020.121040>.
- Renner, S., 2018. Poverty and distributional effects of a carbon tax in Mexico. *Energy Policy* 112, 98–110. <https://doi.org/10.1016/j.enpol.2017.10.011>.
- Rosas-Flores, J.A., Bakhat, M., Rosas-Flores, D., Fernández Zayas, J.L., 2017. Distributional effects of subsidy removal and implementation of carbon taxes in Mexican households. *Energy Econ.* <https://doi.org/10.1016/j.eneco.2016.10.021>.
- Saelim, S., 2019. Carbon tax incidence on household demand: effects on welfare, income inequality and poverty incidence in Thailand. *J. Clean. Prod.* 234, 521–533. <https://doi.org/10.1016/j.jclepro.2019.06.218>.
- Shi, Q., Ren, H., Cai, W., Gao, J., 2019. How to set the proper level of carbon tax in the context of Chinese construction sector? A CGE analysis. *J. Clean. Prod.* 240, 117955. <https://doi.org/10.1016/j.jclepro.2019.117955>.

- Strand, J., 2013. Strategic climate policy with offsets and incomplete abatement: carbon taxes versus cap-and-trade. *J. Environ. Econ. Manage.* 66, 202–218. <https://doi.org/10.1016/j.jeem.2013.03.002>.
- Tan, R., Lin, B., 2020. The influence of carbon tax on the ecological efficiency of China's energy intensive industries—A inter-fuel and inter-factor substitution perspective. *J. Environ. Manage.* 261, 110252. <https://doi.org/10.1016/j.jenvman.2020.110252>.
- The Central People's Government of the People's Republic of China, 2017. National Population Development Plan (2016–2030) [WWW Document]. URL http://www.gov.cn/zhengce/content/2017-01/25/content_5163309.htm (accessed 3.22.20).
- Umit, R., Schaffer, L.M., 2020. Attitudes towards carbon taxes across Europe: the role of perceived uncertainty and self-interest. *Energy Policy* 140, 111385. <https://doi.org/10.1016/j.enpol.2020.111385>.
- Vandeyck, T., Van Regemortel, D., 2014. Distributional and regional economic impact of energy taxes in Belgium. *Energy Policy* 72, 190–203. <https://doi.org/10.1016/j.enpol.2014.04.004>.
- Wang, Q., Hubacek, K., Feng, K., Guo, L., Zhang, K., Xue, J., Liang, Q.-M., 2019. Distributional impact of carbon pricing in Chinese provinces. *Energy Econ* 81, 327–340. <https://doi.org/10.1016/j.eneco.2019.04.003>.
- Wang, Q., Hubacek, K., Feng, K., Wei, Y.-M., Liang, Q.-M., 2016. Distributional effects of carbon taxation. *Appl. Energy* 184, 1123–1131. <https://doi.org/10.1016/j.apenergy.2016.06.083>.
- Xu, B., Lin, B., 2018a. Investigating the role of high-tech industry in reducing China's CO₂ emissions: a regional perspective. *J. Clean. Prod.* 177, 169–177. <https://doi.org/10.1016/j.jclepro.2017.12.174>.
- Xu, B., Lin, B., 2018b. Do we really understand the development of China's new energy industry? *Energy Econ.* 74, 733–745. <https://doi.org/10.1016/j.eneco.2018.07.024>.
- Yamazaki, A., 2017. Jobs and climate policy: evidence from British Columbia's revenue-neutral carbon tax. *J. Environ. Econ. Manage.* 83, 197–216. <https://doi.org/10.1016/j.jjeem.2017.03.003>.
- Yan, Y., Zhang, X., Zhang, J., Li, K., 2020. Emissions trading system (ETS) implementation and its collaborative governance effects on air pollution: the China story. *Energy Policy* 138, 111282. <https://doi.org/10.1016/j.enpol.2020.111282>.
- Zhang, H., Fan, L.-W., 2019. Can emission trading help to improve energy efficiency in China? *Energy Effic.* 12, 979–991. <https://doi.org/10.1007/s12053-018-9735-4>.
- Zhang, H., Hewings, G.J.D., Zheng, X., 2019a. The effects of carbon taxation in China: an analysis based on energy input-output model in hybrid units. *Energy Policy* 128, 223–234. <https://doi.org/10.1016/j.enpol.2018.12.045>.
- Zhang, K., Xue, M.-M., Feng, K., Liang, Q.-M., 2019b. The economic effects of carbon tax on China's provinces. *J. Policy Model.* 41, 784–802. <https://doi.org/10.1016/j.jpolmod.2019.02.014>.
- Zhang, Y.-J., Liang, T., Jin, Y.-L., Shen, B., 2020. The impact of carbon trading on economic output and carbon emissions reduction in China's industrial sectors. *Appl. Energy* 260, 114290. <https://doi.org/10.1016/j.apenergy.2019.114290>.
- Zhijie Jia:** Ph.D. Candidate Co-author E-Mail: zjjia_cn@163.com
- Affiliation: School of Management, China Institute for Studies in Energy Policy, Collaborative Innovation Center for Energy Economics and Energy Policy, Xiamen University, Fujian 361005, PR China
- Ph.D. candidate in Xiamen University, who major in energy economy and environmental economy. The author has published 17 papers, H-index: 8; total citation: 237. His publications are in *Applied Energy*, *Energy Economics*, *Energy Policy*, *China Economic Review*, *Journal of Cleaner Production*, etc.
- Boqiang Lin:** Professor Corresponding author E-Mail: bqlin@xmu.edu.cn, bqlin2004@vip.sina.com
- Affiliation: School of Management, China Institute for Studies in Energy Policy, Collaborative Innovation Center for Energy Economics and Energy Policy, Xiamen University, Fujian 361005, PR China
- Lin Boqiang obtained his Ph.D. in economics from University of California at Santa Barbara. He is a “Chang Jiang Scholar” Professor, Dean of China Institute for Studies in Energy Policy and Director for Collaborative Innovation Center for Energy Economics and Energy Policy at Xiamen University; Vice Chairman of China Energy Society. He is currently also a member of Board of Stewards of Future of Energy of the World Economic Forum based in Davos Switzerland.